

P R A T E E K S I N G H

B.Tech CSE • IIIT Guwahati '27 | Full-Stack & ML Engineer

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GSSoC 2026 Project Admin • GSSoC 2025 Contributor • Hacktoberfest 2025 • CodeSocial 2025

E D U C A T I O N

Indian Institute of Information Technology, Guwahati

2023 - 2027

B.Tech in Computer Science & Engineering

T E C H N I C A L S K I L L S

Languages	Java, Python, C++, JavaScript, TypeScript, HTML, CSS
Frontend	React.js, TailwindCSS, Vite, CSS Modules, Context API
Backend	Node.js, Express.js, Servlet/JSP, REST APIs
Databases	MySQL, MongoDB, Mongoose, Firebase (Auth & Firestore)
AI / ML	PyTorch, Scikit-learn, GANs (Pix2Pix / U-Net / PatchGAN), OpenRouter API, Groq AI
Tools & Platforms	Git, GitHub, VS Code, Linux, Jupyter Notebook, n8n, Ollama, Vercel, JDBC
Libraries	NumPy, Pandas, Matplotlib, Seaborn, Albumentations
Core	OOP, REST APIs, Secure Coding (OWASP), Competitive Programming, Problem Solving, Debugging
Coursework	DSA, OS, DBMS, AI, Computer Architecture, Discrete Mathematics, Optimization Techniques

P R O J E C T S

AuraSpot - AI-Powered Real Estate Marketplace [GitHub](#)

2024

React • TypeScript • Node.js • Express.js • MongoDB • Firebase • DeepSeek AI (OpenRouter)

- ▶ Full-stack real estate platform for property listing, search, and owner-tenant connections, with login via Email/Password or Google.
- ▶ Integrated DeepSeek AI via OpenRouter to score properties, detect suspicious listings, suggest fair rent prices, and surface relevant matches.
- ▶ Built real-time chat, rent agreements, monthly payment tracking, and automated reminders for due and overdue payments.

Sketch to Colorization using Conditional GAN [GitHub](#)

2024

Python • PyTorch • U-Net • PatchGAN • Albumentations • Jupyter Notebook

- ▶ Built a cGAN (Pix2Pix-based) that converts anime-style black-and-white sketches into colored images.
- ▶ Generator uses a U-Net with skip connections to preserve fine sketch details; Discriminator is a PatchGAN that evaluates image patches for sharper textures.
- ▶ Trained on the Anime Sketch Colorization Pair dataset. Full workflow - data loading, training, inference - contained in a single notebook.

CodeCapsule - Offline AI Coding Assistant (VS Code Extension) [GitHub](#)

2025

JavaScript • VS Code Extension API • Ollama • Local LLMs

- ▶ VS Code extension that runs entirely offline using local models via Ollama - no internet, no data sent out.
- ▶ Supports code explanation, bug detection, refactoring suggestions, and code generation, all within the editor.

Automated Documentation Agent [GitHub](#)

2026

n8n • Groq AI • Webhook Triggers • REST APIs

- ▶ Workflow automation agent built in n8n that uses Groq AI to automatically generate documentation for codebases on trigger.
- ▶ Demonstrates agentic automation - chaining AI inference, webhook handling, and data transformation without custom backend code.

The Matrix - Gamified Learning OS [GitHub](#)

2026

JavaScript • HTML • CSS • Node.js

- Interactive gamified environment designed to make learning programming concepts engaging through simulated OS-like challenges.

EXPERIENCE & OPEN SOURCE

Project Admin - GSSoC 2026 (GirlScript Summer of Code)

2026

- Managing contributor onboarding, PR reviews, issue triaging, and project health for an open-source project under the GSSoC 2026 program.
- Responsible for maintaining code quality standards, mentoring contributors, and ensuring timely merges.

Research Collaborator - KathiraveluLab, University of Alaska Anchorage

2025 - 2026

- Contributed to [GUARDA](#) (Gateway for Uniform Access to Remote Data & Analytics) - a federated database query platform - under Dr. Pradeeban Kathiravelu's research group.
- Received an official reference letter from Dr. Kathiravelu (Assistant Professor of CS, UAA) recognising contributions to research and open-source collaboration.
- Work involved secure coding practices, issue tracking, code review, and remote collaboration across time zones.

Open Source Contributor - GSSoC 2025, Hacktoberfest 2025, CodeSocial 2025

2025

- Merged multiple PRs across different repositories - bug fixes, code cleanup, documentation updates, and feature additions.
- Reviewed code for security issues, followed OWASP guidelines, and collaborated globally via Git, GitHub Issues, and PR workflows.
- Comfortable with real open-source workflows: branching, rebasing, CI checks, and clear commit messages.